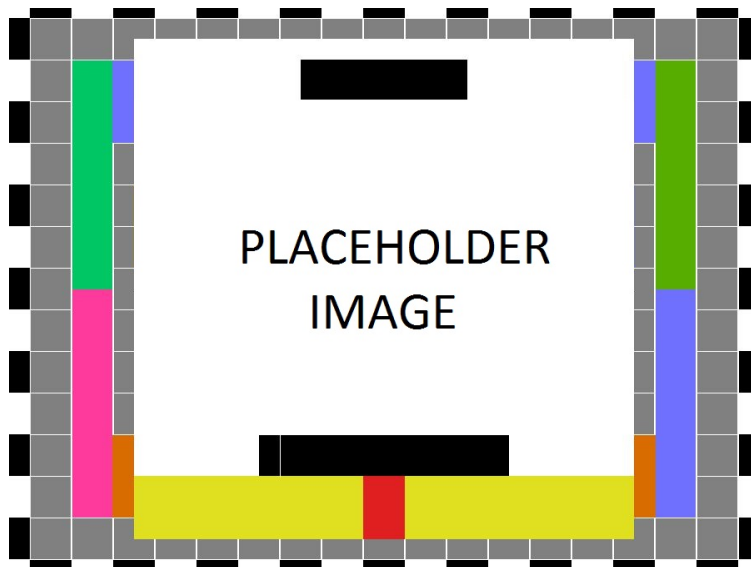


**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**PROJECT CHARTER
CSE 4316: SENIOR DESIGN I
FALL 2021**



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REVISION HISTORY

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1.0	10.20.2021	AT, GH, CB	official release
1.1	10.31.2021	AL	added customer change requests

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1 PROBLEM STATEMENT

The Senior Design labs contain a large inventory of equipment and other items that are frequently used by students. However, many students are unfamiliar with the lab and the locations of many items. As a result, the lab manager receives many inquiries about where different items are stored. These inquiries both slow down the progress of students as they await a response, and distract the lab manager from other responsibilities. Moreover, many creative projects created in these labs remain unknown to students and others outside of the lab.

2 METHODOLOGY

We will build a website that manages lab inventory and provides an accessible way for lab users to locate equipment. Users will be able to search for items/equipment, even when they do not know its exact name or terminology. An item's info page will describe its location and appearance. We will also utilize speech-to-text recognition so users can easily search up items. The website will provide a space to showcase creative projects created in the lab.

3 VALUE PROPOSITION

The Value Proposition explains how the sponsors will benefit from your work, and why they should invest funding, time, and expertise in supporting your team. Here, you are essentially making a case for the project. There are many ways in which value can be returned to your stakeholders (industrial sponsors, instructors, the university, etc.), list any that may help you convince them to "buy in".

The 3D Lab Manager gives Lab 208 in the Engineering Research Building a single system that shows what equipment exists, where it sits on a 3D scan of the lab, and whether it is available right now. A touchscreen kiosk in the lab puts this information in front of students when they need it. A previous team attempted this system but never completed it.

The project returns value to its stakeholders in many ways:

- **Time returned to the lab manager.** Students can find items on their own instead of contacting Steven McDermott, freeing him for his other responsibilities.
- **Protection of high value equipment.** Wireless tracking shows where expensive tools are, reducing loss and protecting the department's investment.
- **Recognition for student work.** A project showcase highlights creative work built in Lab 208 for prospective students to see and credits the students who made it.
- **Promotion of UTA.** Public 3D scans of UTA labs let prospective students, sponsors, and visitors explore the university's facilities.
- **Long term value.** The system runs on university servers and keeps serving future senior design teams at no licensing cost.

4 DEVELOPMENT MILESTONES

This list of core project milestones should include all major documents, demonstration of major project features, and associated deadlines. Any date that has not yet been officially scheduled at the time of preparing this document may be listed by month.

Provide a list of milestones and completion dates in the following format:

- Project Charter first draft: September 2026

- System Requirements Specification: October 2026
- Architectural Design Specification: November 2026
- Demonstration of website prototype with inventory database and basic search: November 2026
- Gate Review presentation: November 2026
- Detailed Design Specification: February 2027
- Demonstration of 3D scan of Lab 208 on the website: February 2027
- Demonstration of improved search with typo handling, search by function, and voice search: February 2027
- Demonstration of touchscreen kiosk: March 2027
- Demonstration of wireless tracking of high value tools: March 2027
- College of Engineering Innovation Day poster presentation: April 2027
- Demonstration of student project showcase and scans of additional UTA labs: April 2027
- Final Project Demonstration: May 2027

5 BACKGROUND

The 208 lab in the ERB, managed by Steven McDermott, contains a large inventory of equipment and other items that are frequently used by students. However, many students are unfamiliar with the lab and the locations of stored equipment. As a result, students often contact Steven McDermott to determine where specific items are located. These inquiries slow down students as they wait for a response and require the lab manager to repeatedly answer questions that could otherwise be resolved independently. This takes time away from Steven McDermott's other responsibilities and is inefficient for accessing information about the lab's inventory.

The current system also makes it difficult for students to find equipment when they do not know its exact name or terminology. For example, a student may know the functionality of an item without knowing what the item is called, or they may enter a search containing a typographical error. A system that relies on users already knowing the correct terminology limits how effectively students can locate equipment. There is an opportunity to improve this process by providing students with a centralized and accessible way to search the lab's inventory.

Steven McDermott has sponsored our team to develop a website that addresses these issues. The website will allow users to search for equipment and other items in the lab and find information about each item, including its location and appearance. The search functionality will be designed to mitigate common user errors, such as typos and searches based on an item's functionality. Speech-to-text functionality will also allow users to search using their voice, providing a convenient method of accessing the inventory.

In addition to improving access to equipment, the project provides an opportunity to increase awareness of the creative projects developed in Lab 208. Many of these projects are currently unknown to students and others outside of the lab. Providing a space to showcase these projects can help inspire other students, demonstrate the work being done in the lab, and give appropriate recognition to the students who created them.

6 RELATED WORK

Discuss the state-of-the-art with respect to your product. What solutions currently exist, and in what form (academic research, enthusiast prototype, commercially available, etc.)? Include references and citations as necessary using the *cite* command, like this [?]. If there are existing solutions, why won't they work for your customer (too expensive, not fast enough, not reliable enough, etc.). This section should occupy 1/2 - 1 full page, and should include at least 5 references to related work. All references should be added to the *.bib* file, fully documented in IEEE format, and should appear in the *references* section at the end of this document (the IEEE citation style will automatically be applied if your reference is properly added to the *.bib* file).

ProTip: Consider using a citation manager such as Mendeley, Zotero, or EndNote to generate your *.bib* file and maintain documentation references throughout the life cycle of the project.

7 SYSTEM OVERVIEW

Explain, at a high level, how you will implement a solution to the problem. Include a diagram of major components to the system (not a full architectural design, but a high level overview of the major system components and how a user or external system might interface). Avoid specific implementation details (operating system, programming languages, etc.). This section should occupy at least 1 full page.

8 ROLES & RESPONSIBILITIES

Who are the stakeholders of the project? Who will be the point of contact from the sponsor or customer side? Who are the team members, and what will be their areas of responsibility? Will your team main-

tain the product owner and scrum master for the whole project, or will that role change periodically? This section should occupy 1/2 - 1 full page.

9 COST PROPOSAL

This section contains the approximate budget for the project, where that money will come from, and any other support. This text should be replaced with a discussion and justification of major expenses, but not the actual monetary amounts (that will go in the preliminary budget section below).

9.1 PRELIMINARY BUDGET

Include a high level budget table for components, fabrication, software licensees, development hardware, etc. This should be in a tabular format broken up into appropriate line items.

9.2 CURRENT & PENDING SUPPORT

What are all of the funding sources for the project, and are there any potential funding sources that haven't been secured yet? List all funding sources (including the default funding amount provided by the CSE department) and their dollar amounts.

10 FACILITIES & EQUIPMENT

What lab space, testing grounds, makerspaces, etc. will you need to complete the project? Will you require any specific equipment, and if so, where will you get it (borrow, lease, purchase, outsource, already present in the lab, etc.). This section should occupy 1/2 page.

11 ASSUMPTIONS

An assumption is a belief of what you assume to be true in the future. You make assumptions based on your knowledge, experience or the information available on hand. These are anticipated events or circumstances that are expected to occur during your project's life cycle.

Assumptions are supposed to be true but do not necessarily end up being true. Sometimes they may turn out to be false, which can affect your project significantly. They add risks to the project because they may or may not be true. For example, if you are working on an outdoor unmanned vehicle, are you assuming that testing space will be available when needed? Are you relying on an external team or contractor to provide a certain subsystem on time? If you are working at a customer facility or deploying on their computing infrastructure, are you assuming you will be granted physical access or network credentials?

This section should contain a list of at least 5 of the most critical assumptions related to your project. For example:

The following list contains critical assumptions related to the implementation and testing of the project.

- Steven McDermott will be available to review progress each sprint cycle
- Steven McDermott will provide a lab inventory list with item names and locations by the 2nd sprint cycle
- The university will provide a server to host the website and database by the 3rd sprint cycle
- The website will be permitted to sign users in with their UTA accounts
- A 3D scanner capable of capturing a full room will be available through purchase or university providing it

- The team will be granted access to scan Lab 208 and other UTA labs
- Wall space, power, and a network connection will be available for the kiosk inside Lab 208

12 CONSTRAINTS

Constraints are limitations imposed on the project, such as the limitation of cost, schedule, or resources, and you have to work within the boundaries restricted by these constraints. All projects have constraints, which are defined and identified at the beginning of the project.

Constraints are outside of your control. They are imposed upon you by your client, organization, government regulations, availability of resources, etc. Occasionally, identified constraints turn out to be false. This is often beneficial to the development team, since it removes items that could potentially affect progress.

This section should contain a list of at least 5 of the most critical constraints related to your project. For example:

The following list contains key constraints related to the implementation and testing of the project.

- Final project demonstration must be completed by May 2027
- Total development costs, including the kiosk tablet, trackers, and scanning equipment, must not exceed the budget
- All purchases must go through the department's SharePoint purchasing process
- The system must be hosted on university servers and follow university security policies
- Lab 208 is shared with other teams, so scanning and testing must not disrupt their work
- Other labs and student projects may only be featured with permission from the people responsible for them

13 RISKS

The following high-level risk census contains identified project risks with the highest exposure. Mitigation strategies will be discussed in future planning sessions.

Risk description	Probability	Loss (days)	Exposure (days)
Inventory information is incomplete or inaccurate	0.40	10	4.0
Search functionality is more difficult to implement than expected	0.35	10	3.5
Changes to project requirements during development	0.30	10	3.0
Speech-to-text functionality requires additional development time	0.30	8	2.4
Database or inventory changes cause unexpected website issues	0.25	8	2.0
Accessibility issues are discovered during testing	0.25	7	1.75

Table 1: Overview of highest exposure project risks

14 DOCUMENTATION & REPORTING

14.1 MAJOR DOCUMENTATION DELIVERABLES

All major documentation deliverables will be versioned at minimum with a revision history table in each document. Revision numbering will begin at 0.1 and increment by 0.1. A major revision may be designated and will begin by finding the ceiling of the current version number. Incremental changes will be tracked through git or another tool integrated into an editor.

14.1.1 PROJECT CHARTER

There will be two milestone releases for the project charter: a first draft at the end of sprint one (Sep 28, 2026) and a final draft at the end of 8. This document should not need frequent revision, however will be updated as deemed necessary by group consensus.

14.1.2 SYSTEM REQUIREMENTS SPECIFICATION

The initial version of the System Requirements Specification will be completed by the end of sprint two (Oct 19, 2026). This document will codify requirements gathered from relevant stakeholders during sprint one as recorded in our requirements spreadsheet. Most of this document should remain static, however customer requirements may move throughout the project as a result of user testing. Substantial changes in scope or features will be updated during the sprint review/planning stage following stakeholder consultation. The final revision of this document will be submitted with the completed project in April of 2027.

14.1.3 ARCHITECTURAL DESIGN SPECIFICATION

The Architectural Design Specification will be completed at the end of sprint three (Nov 9, 2026). Barring major changes to requirements or hardware availability, we do not anticipate substantial modifications to the architectural documentation. In the event of a architectural redesign, relevant decisions will be reflected in this document. The final iteration will be submitted in April of 2027.

14.1.4 DETAILED DESIGN SPECIFICATION

The Detailed Design Specification will be completed with the wrap up of Sprint 5 in early February. As implementation progresses in Senior Design 2, implementation details may change substantially. Following the completion of successive sprints the team will compare implementation details to the documentation and ammend any discrepancies. The final copy of this document will be submitted in April of 2027 with the completed project.

14.2 RECURRING SPRINT ITEMS

14.2.1 PRODUCT BACKLOG

Features will be added to the product backlog in consultation with the project owner and the System Requirements Specification. Changes proposed by stakeholders will ultimately be voted upon by the development team before adoption. The product backlog will be tracked by the project Jira board.

14.2.2 SPRINT PLANNING

There will be a total of eight sprints across both semesters, with one elapsed from the first revision of this document. Planning for the next sprint will occur in discussion with the project owner during the end of the previous sprint.

14.2.3 SPRINT GOAL

During the sprint review stakeholder consultation the development team shall elicit a sprint goal from the product owner. Failing a clear directive from the owner, the team shall identify necessary technical debt to remove to improve reliability.

14.2.4 SPRINT BACKLOG

After the determination of the sprint goal, the group shall discuss which features most directly accomplish the goal and are achievable within the available timeframe. The sprint backlog will be tracked in a Jira scrum board alongside the product backlog.

14.2.5 TASK BREAKDOWN

Group members will first volunteer for tasks, remaining tasks will be reasonably divided amongst team members which did not volunteer such that the overall assignment of work is roughly equal.

14.2.6 SPRINT BURN DOWN CHARTS

Liam will be responsible for creating the sprint burn down chart at the end of each sprint. This is contingent on team members tracking time and item completion in Jira. The burndown chart will be produced by Jira similar to the one below.

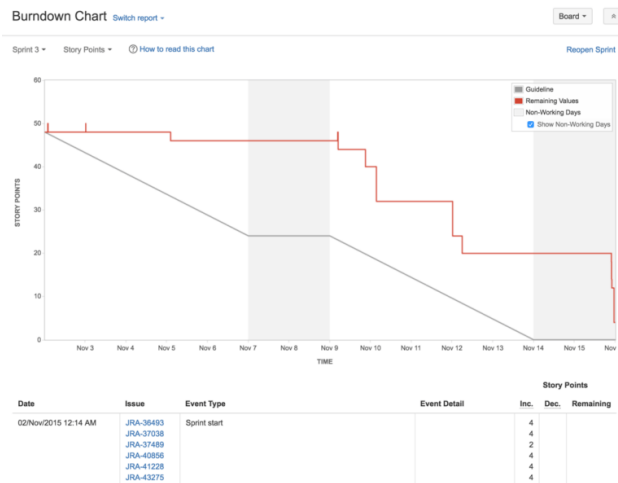


Figure 1: Example sprint burn down chart

14.2.7 SPRINT RETROSPECTIVE

The sprint retrospective will occur during the creation of the sprint review slides. Reporting will take the form of the provided template, documenting (among other things) the: goal, completed items, and burndown chart. Individual sprint reviews are to be completed by team members on their own time frame, before the due date.

14.2.8 INDIVIDUAL STATUS REPORTS

What sort of status will be reported by each individual member, and how often will it be reported? What key items will be contained in the report?

14.2.9 ENGINEERING NOTEBOOKS

How often will the engineering notebook be updated, at a minimum, by each team member? What is the minimum amount of pages that will be completed for each interval, and how long will that interval be? How will the team keep each member accountable? Who will sign off as a "witness" for each ENB page?

14.3 CLOSEOUT MATERIALS

The following materials, in addition to major documentation deliverables, will be provided to the customer upon project closeout. Remove this paragraph from your draft, but leave the heading.

14.3.1 SYSTEM PROTOTYPE

What will be included in the final system prototype? How and when will this be demonstrated? Will there be a Prototype Acceptance Test (PAT) with your customer? Will anything be demonstrated off-site?

If so, will there be a Field Acceptance Test (FAT)?

14.3.2 PROJECT POSTER

What will be included on the poster, what will be the final dimensions, and when will it be delivered?

14.3.3 WEB PAGE

What will be included on the project web page? Will it be accessible to the public? When will this be delivered? Will it be updated throughout the project, or just provided at closeout (at a minimum, you need to provide a simple web page at the end).

14.3.4 DEMO VIDEO

What will be shown in the demo video(s)? Will you include a B-reel footage for future video cuts? Approximately how long will the video(s) be, and what topics will be covered?

14.3.5 SOURCE CODE

How will your source code be maintained? What version control system will you adopt? Will source code be provided to the customer, or binaries only? If source code is provided, how will it be turned over to the customer? Will the project be open sourced to the general public? If so, what are the license terms (GNU, GPL, MIT, etc.). Where will the license terms be listed (in each source file, in a single readme file, etc.).

14.3.6 SOURCE CODE DOCUMENTATION

What documentation standards will be employed? Will you use tools to generate the documentation (Doxygen, Javadocs, etc.). In what format will the final documentation be provided (PDF, browsable HTML, etc.)?

14.3.7 HARDWARE SCHEMATICS

Will you be creating printed circuit boards (PCBs) or wiring components together? If so, list each applicable schematic and what sort of data it will contain (PCB layout, wiring diagram, etc.). If your project is purely software, omit this section.

14.3.8 CAD FILES

Will the project involve any mechanical design, such as 3D printed or laser-cut parts? If so, what software will you use to generate the files and what file formats will you provide in your closeout materials (STL, STEP, OBJ, etc.). If your project is purely software, omit this section.

14.3.9 INSTALLATION SCRIPTS

How will the customer deploy software to new installations? Will you provide installation scripts, install programs, or any other tools to improve the process? Will there be multiple scripts provided (perhaps separate scripts for the graphical front end and back end server software)?

14.3.10 USER MANUAL

Will your customer need a printed or digital user manual? Will they need a setup video? Decide now what will be provided and discuss.